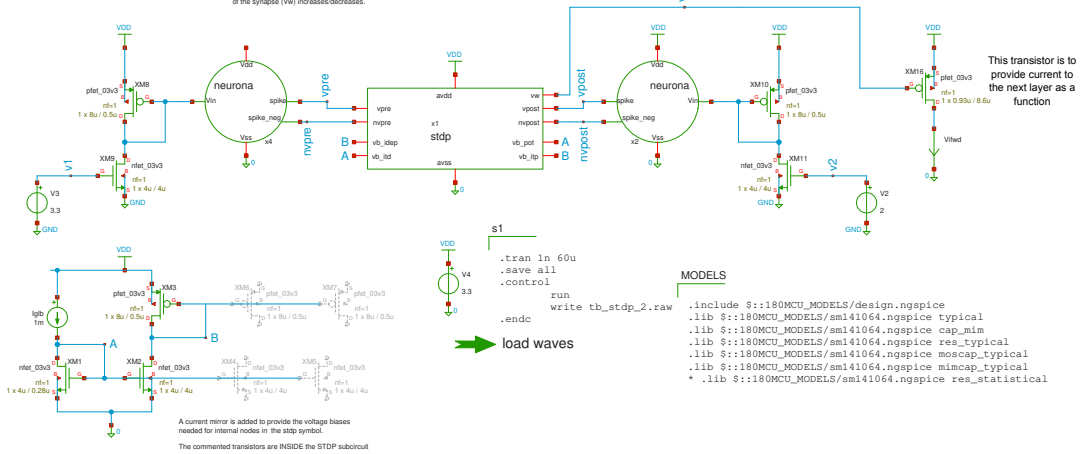


TB: two neurons (one presynaptic [left] and the other post-synaptic [right]) are receiving a ramp of voltage each provide spikes (and its complementary signal) to a synapse. Each time the presynaptic/postsynaptic neuron spike, the weight value of the synapse (Vw) increases/decreases.



```

s1
.tran in 60u
.save all
.control
run
write tb_stdp_2.raw
.endc

```

MODELS

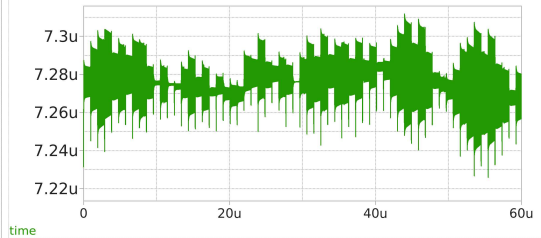
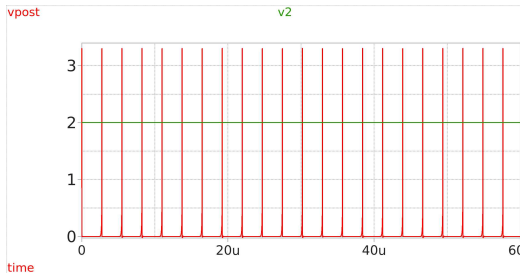
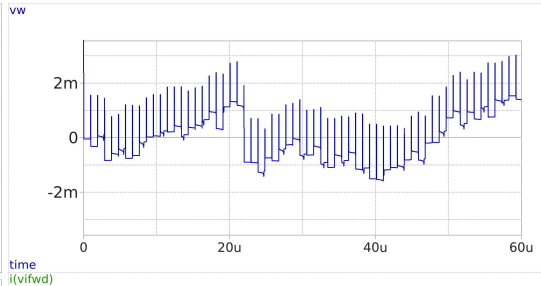
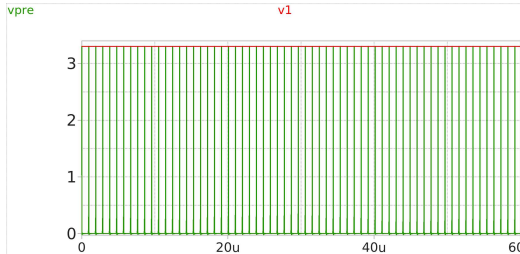
```

.include $::180MCU_MODELS/design.ngspice
.lib $::180MCU_MODELS/sml41064.ngspice typical
.lib $::180MCU_MODELS/sml41064.ngspice cap_mim
.lib $::180MCU_MODELS/sml41064.ngspice res_typical
.lib $::180MCU_MODELS/sml41064.ngspice moscap_typical
.lib $::180MCU_MODELS/sml41064.ngspice mimcap_typical
* .lib $::180MCU_MODELS/sml41064.ngspice res_statistical

```

The spikes of the presynaptic neuron. On purpose, it is set to spike faster than the post synaptic neuron.

As the presynaptic neuron spikes faster, the synaptic value Vw grows



The spikes of the postsynaptic neuron. On purpose, it is set to spike faster than the post synaptic neuron.

The spikes of the postsynaptic neuron. On purpose, it is set to spike faster than the post synaptic neuron.